

Albert J. Lee

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RESEARCH SUMMARY

ML Ph.D. candidate moving into **robot learning for humanoids**, and quick to become productive in a new field. I joined PIA Robotics, a pre-seed humanoid startup, with no prior robotics experience and within two months built its **full sim-to-real pipeline for humanoid whole-body control** from scratch: RL training in Isaac Lab, sim-to-sim validation in MuJoCo, and deployment on a real humanoid. My current work turns text-generated human motion into robot reference data through **mesh-based retargeting**, with the goal of making the data clean enough that RL is needed as little as possible. My Ph.D. work used causal interventions on model internals (activation patching, steering, targeted post-training) to find where specific behaviors come from in LLMs and VLMs; before that, I trained in neuroscience. Seeking research roles in humanoid loco-manipulation and simulation.

EDUCATION

Ph.D. in Machine Learning, Georgia Institute of Technology

Anticipated 12/2026

B.S. in Neuroscience (Faculty Honors), Georgia Institute of Technology

RESEARCH EXPERIENCE

Research Intern, PIA Robotics

08/2026–present

Pre-seed humanoid startup · whole-body control and sim-to-real (Unitree G1)

- Built the company's **end-to-end sim-to-real pipeline** from scratch: RL training of whole-body control policies in Isaac Lab, sim-to-sim validation in MuJoCo, and deployment on hardware.
- Building a **reference-motion pipeline** that generates human motion from text with a kinematic motion-diffusion model, converts it to a parametric body mesh, and retargets it to the robot through dense body-surface correspondence under contact, joint-limit, and floor constraints, so that the reference data is clean enough for RL to be needed only minimally.
- Policies from the pipeline run in the company's **teleoperated hardware demo**, in which the robot walks through doorways, turns, and crouches while holding an object in both hands.

Research Intern, LG AI Research (Bio Intelligence Lab)

05/2026–09/2026

- Developed **DCCD-J**, a differentiable causal discovery method for interventional data under latent confounding and feedback loops; confounder-recovery F1 0.93 vs. 0.23 for the prior method under dense confounding. Co-first author, **NeurIPS 2026 Workshop on AI for Drug Discovery (AI4DD)**.

Graduate Research Assistant (PI: C. S. Mitchell, Ph.D.)

08/2021–present

Laboratory for Pathology Dynamics, Georgia Tech & Emory University School of Medicine

- **Robustness fine-tuning (ACL ARR 2026 submission)**: showed that KL-regularized LoRA fine-tuning of Llama-3.1-8B (vs. GRPO) to resist wrong diagnostic suggestions trades accuracy under correct ones (89% → 62%) for accuracy under wrong ones (31% → 72%); localized ~60% of the change to one layer and partly reversed it.
- **Component-restricted post-training (ongoing)**: localized an LLM's rule-application failure to about 2% of attention heads and post-trained only those heads (SFT + RL with verifiable rewards), comparing against budget-matched random-head and full-parameter baselines on correction vs. side effects.
- **Causal auditing of VLMs (AAAI 2027 submission)**: showed that activation patching that repairs a VLM's answer need not restore evidence from the image; different-case donors reproduced 98–103% of same-case recovery across 5 VLMs.
- **Shared bias direction (ACL ARR 2026 submission)**: one activation direction reduced cue-following across seven medical QA cue types and three LLM families.
- **HiCo-EL (Findings of EMNLP 2026)**: hierarchical contrastive pretraining for bi-encoder entity linking; +3.7% R@1 on MedMentions and +9.6% R@1 on unseen NCBI Disease entities; supervised two junior researchers.
- **Earlier work (2021–2025)**: contributed to the lab's biomedical text-mining tools (CCS Explorer, TrialSieve, SemNet Explorer), with HiCo-EL as the entity-linking step; interpretable ML for cognitive assessment (first author, J. Clin. Med. 2025).

Undergraduate Research Fellow and ALS Team Lead (PI: C. S. Mitchell, Ph.D.)

12/2016–05/2020

- Led an NIH grant and a team of 5 researchers on data-driven meta-analysis and dynamical-systems modeling of ALS, resulting in 9 publications.

SKILLS

- **Robotics:** Isaac Lab, MuJoCo; RL for humanoid whole-body control; sim-to-sim and sim-to-real transfer on a Unitree G1; text-to-motion diffusion models and mesh-based motion retargeting.
- **Machine learning:** Python, PyTorch, HuggingFace, Accelerate, DeepSpeed, multi-GPU training, W&B, MATLAB, R; SFT, GRPO (RL with verifiable rewards), LoRA; activation patching and causal interventions; controlled evaluation design.
- **Languages:** Native in Korean; fluent in English.

PUBLICATIONS

Under review:

- [1] **Lee, A. J. B.**, & Mitchell, C. S. Recovery Is Not Provenance: A Source-Substitution Audit of VLM Activation Patching in Medical VQA. Under review at AAAI 2027.
- [2] **Lee, A. J. B.**, & Mitchell, C. S. Seven Medical Reasoning Biases, One Causal Direction. Under review at ACL Rolling Review (2026).
- [3] **Lee, A. J. B.**, & Mitchell, C. S. Resisting Wrong Suggestions, Losing Right Ones: What Robustness Fine-Tuning Changes in Medical LLMs. Under review at ACL Rolling Review (2026).

Peer-reviewed conference papers:

- [4] **Lee, A. J. B.**, Bathala, P., Malhotra, P., Ye, C., & Mitchell, C. S. (2026). HiCo-EL: Encoding Ontology Structure into Bi-Encoder Entity Linking. Findings of EMNLP 2026.
- [5] Al-Hussaini, I., An, D. N., **Lee, A. J.**, Bi, S., & Mitchell, C. S. (2022). CCS Explorer: Relevance Prediction, Extractive Summarization, and Named Entity Recognition from Clinical Cohort Studies. IEEE International Conference on Big Data (IEEE BigData 2022), Oral.

Peer-reviewed workshop papers:

- [6] **Lee, A. J. B.***, Kang, J.*, Kim, S.*, & Jeong, H. (2026). Separating Co-expression from Regulation in Cyclic Causal Discovery with Latent Confounders. NeurIPS 2026 Workshop on AI for Drug Discovery (AI4DD). *Equal contribution.

Peer-reviewed journal articles:

- [7] He, X., Camacho, D., Moukheiber, L., Iyer, M., Zhao, B., Ye, C., Nursal, B., Guo, X., **Lee, A. J. B.**, & Mitchell, C. S. (2026). SemNet Explorer: An Evidence-Grounded Knowledge Graph-LLM Framework for Multi-Scale Mechanistic Reporting Across Biomedical Domains. Big Data and Cognitive Computing, 10(6), 171.
- [8] Fay, M. E., Tandon, R., Latham, T., **Lee, A. J.**, Rankine-Mullins, A. E., Reid, M., Mitchell, C. S., Ware, R. E., & Lam, W. A. (2026). Artificial Intelligence-Based Analysis of Central Nervous System Vasculopathy in Pediatric Sickle Cell Anemia. American Journal of Hematology, 101(5), 998–1004.
- [9] **Lee, A. J. B.**, Zhao, B., Lah, J. J., John, S. E., Loring, D. W., & Mitchell, C. S. (2025). Education, Sex, and Age Shape Rey Complex Figure Performance in Cognitively Normal Adults: An Interpretable Machine Learning Study. Journal of Clinical Medicine, 14(21), 7562.
- [10] Kartchner, D., Turner, H., Ye, C., Al-Hussaini, I., Nursal, B., **Lee, A. J. B.**, Deng, J., Curtis, C., Cho, H., Duvaris, E. L., Jackson, C., Shanks, C. E., Tan, S. Y., Ramalingam, S., & Mitchell, C. S. (2025). TrialSieve: A Comprehensive Biomedical Information Extraction Framework for PICO, Meta-Analysis, and Drug Repurposing. Bioengineering, 12(5), 486.
- [11] **Lee, A. J. B.**, Bi, S., Ridgeway, E., Al-Hussaini, I., Deshpande, S., Krueger, A., Khatri, A., Tsui, D., Deng, J., & Mitchell, C. S. (2025). Restoring Homeostasis: Treating Amyotrophic Lateral Sclerosis by Resolving Dynamic Regulatory Instability. International Journal of Molecular Sciences, 26(3), 872.
- [12] **Lee, A. J. B.**, Kittel, T. E., Kim, R. B., Bach, T. N., Zhang, T., & Mitchell, C. S. (2022). Comparing Therapeutic Modulators of the SOD1 G93A Amyotrophic Lateral Sclerosis Mouse Pathophysiology. Frontiers in Neuroscience, 16, 1111763.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Georgia Tech, Dept. of Biomedical Engineering

Summer 2025

- Graded homework, held weekly office hours, and reviewed concepts for BMED2400 (Bioengineering Statistics).

HONORS / FELLOWSHIPS

BME Department Outstanding Teaching & Mentorship Award — Nominee	2025
National Science Foundation Graduate Research Fellowship — Honorable Mention	05/2020
NIH T32 Training Fellowship in Computational Neuroengineering	05/2020
President's Undergraduate Research Project Award	
President's Undergraduate Research Travel Award	10/2017
Zell Miller Scholarship (full-tuition merit-based scholarship)	